

Empirical rule with z-scores



1 -2

2 3

3

a $x = 65\%$

b $x = 68\%$

4

a 0.13

b 147

5 Mathematics

6 English

7

a 2.7

b 3.7

c Gwen

8

a -1.2

b -2.9

c Mathematics

9

a 0.9265 units^2

b 0.0570 units^2

c 0.1093 units^2

d 0.8790 units^2

e 0.0336 units^2

f 0.0809 units^2

10

a 93.45%

b 86.40%

11

a 0.1593

b 0.6607

c 0.9274

d 0.8527

Calculations with the standard and general normal distributions

12

a 0.8528

b -0.8183

c -0.7790

d 1.0102

e 0.8638

f 1.0141

13 a 21.93 b 20.63  c 10.54 d 30.19

14 a 3 b 91 c 2.4 d 76.8

Quantiles and percentiles

15 a 0.524 b 0.385 c 0.842

16 a 0 b 1 c -1

17 a 45 b 40 c 60

18 a 41 cm b 37 cm

19 a 91.584 b 112.816 c 94.756

20 a 30 b 33.82 c 28.46

21 a 1.600 b -0.385 c 48.075

22 To be accepted into Mensa, a person must score higher than 98% of other people on the IQ test or a successful applicant's IQ score will be among the top 2% of the population.

Unknown mean or standard deviation

23 46

24 60



26 6.5

27 11.54

28 16.02

29 $\mu = 73.33, \sigma = 3.67$

30 $\mu = 14.02, \sigma = 3.57$